University of Cape Town

Department of Mathematics and Applied Mathematics

Mathematics II **Advanced Calculus (2AC)**

Class Test 1 **22nd March 2023**

Time: 1.5 hrs **Full Mark: 37**

Solutions:

Section A **Full Mark: 27**

In this section full answers are expected. Marks will be deducted for incomplete solutions.

1. For each of the functions below say whether the limit exists at the point $(0,0)$. If the limit exists, then find it.

(a) $f_1(x,y) := \dfrac{x^2\sin x - y\sin y + y\sin x - x^2\sin y}{y + x^2}$;

We notice that

$$\lim_{(x,y)\to(0,0)} \left[x^2\sin x - y\sin y + y\sin x - x^2\sin y\right] = 0 \quad \text{and} \quad \lim_{(x,y)\to(0,0)} (y+x^2) = 0\,.$$

We also notice that the terms $\sin x$ and $\sin y$ are repeated and hence, one can consider some factorization in the numerator as follows:

$$\begin{aligned} x^2\sin x - y\sin y + y\sin x - x^2\sin y &= [x^2\sin x + y\sin x] + [-y\sin y - x^2\sin y] \\ &= (x^2+y)\sin x - (y+x^2)\sin y \\ &= (x^2+y)(\sin x - \sin y)\,. \checkmark \end{aligned}$$

Thus,

$$\begin{aligned} \lim_{(x,y)\to(0,0)} f_1(x,y) &= \lim_{(x,y)\to(0,0)} \frac{(x^2+y)(\sin x - \sin y)}{y+x^2} \\ &= \lim_{(x,y)\to(0,0)} (\sin x - \sin y) = 0\,. \checkmark \end{aligned}$$

[2]

(b) $f_2(x,y) := \dfrac{x\ln(1+|y|)}{\sqrt{|x|^3+y^2}}$, (Hint: $\ln(1+t) \le t$ for every $t \ge 0$);

First solution: We have $f_2(x,0) = 0$ and hence the limit along the x-axis is equal to 0. (1/2 ✓) Similarly, also the limit along the y-axis is equal 0. (1/2 ✓). So, the possible candidate as limit would be $L = 0$. We can use the hint as follows:

$$\begin{aligned} |f_2(x,y) - 0| &= \frac{|x|\ln(1+|y|)}{\sqrt{|x|^3+y^2}} \le \frac{|x|\,|y|}{\sqrt{|x|^3+y^2}} \checkmark \quad \text{(Using the hint } \ln(1+|y|) \le |y|) \\ &= |x| \times \underbrace{\left[\frac{|y|}{\sqrt{|x|^3+y^2}}\right]}_{\le 1} \le |x| \ (1/2 \checkmark) \quad \longrightarrow 0 \quad \text{as } (x,y)\to(0,0). \end{aligned}$$

Hence, by the squeeze theorem, we obtain that $\lim_{(x,y)\to(0,0)} f_2(x,y) = 0$. (1/2 ✓)

Second solution: We could also write $f_2(x,y) = \dfrac{\ln(1+|y|)}{|y|} \times \dfrac{x|y|}{\sqrt{|x|^3+y^2}}$. (1/2 ✓) Now, using $\lim_{t\to 0} \dfrac{\ln(1+t)}{t} = 1$, (1/2 ✓) we only need to find the limit

$$\lim_{(x,y)\to(0,0)} \frac{x|y|}{\sqrt{|x|^3+y^2}} = 0 \text{ (as done above)} \checkmark + \text{(1/2 ✓)}$$

Hence, $\lim_{(x,y)\to(0,0)} f_2(x,y) = 0$. (1/2 ✓)

Third solution: We could also use polar coordinates as follows: knowing that the candidate is equal to 0. (1/2 ✓) as above, we set $x = r\cos\theta$, $y = r\sin\theta$ and get

$$\begin{aligned}
|f_2(r\cos\theta,\, r\sin\theta) - 0| = \frac{|r\cos\theta|\,\ln(1+r|\sin\theta|)}{\sqrt{r^2(r|\cos\theta|^3+\sin^2\theta)}} &= \frac{|\cos\theta|\,\ln(1+r|\sin\theta|)}{\sqrt{r|\cos\theta|^3+\sin^2\theta}}. \text{ (1/2 ✓)}\\
&\le \frac{|\cos\theta|\,r|\sin\theta|}{\sqrt{r|\cos\theta|^3+\sin^2\theta}} \checkmark \text{ (using the lhint)}\\
&\le \frac{r|\sin\theta|}{|\sin\theta|} = r \to 0 \text{ (1/2 ✓) as } r\to 0^+ .
\end{aligned}$$

Hence, $\lim_{(x,y)\to(0,0)} f_2(x,y) = 0$. (1/2 ✓)

[3]

(c) $f_3(x,y) := \dfrac{y^2}{x^2}\sin\Big(\dfrac{x}{\sqrt{x^2+y^2}}\Big)$;

We notice that along the x-axis ($y=0$), we get $f_3(x,0) = 0$ and hence, the along that axis is equal to 0. (1/2 ✓)
We cannot consider the y-axis here. Let us consider for instance the line $y = x$ and get

$$f_3(x,x) = \frac{x^2}{x^2}\sin\Big(\frac{x}{\sqrt{2x^2}}\Big) = \sin\Big(\frac{x}{|x|\sqrt{2}}\Big) = \begin{cases} \sin\Big(\dfrac{1}{\sqrt{2}}\Big) \neq 0 & \text{if } x > 0, \\ \sin\Big(-\dfrac{1}{\sqrt{2}}\Big) \neq 0 & \text{if } x < 0\,. \end{cases} \quad \text{(1/2 ✓)}$$

Thus,

$$\lim_{x\to 0^+} f_3(x,x) = \sin\Big(\frac{1}{\sqrt{2}}\Big) \neq \sin\Big(-\frac{1}{\sqrt{2}}\Big) = \lim_{x\to 0^-} f_3(x,x)\,. \text{ (1/2 ✓)}$$

So, the limit does not even exist along the line $y = x$ and hence, $\lim_{(x,y)\to(0,0)} f_3(x,y)$ does not exist. (1/2 ✓)

[2]

2. Consider the vector function given by $\vec{r}(t) := \Big(\dfrac{\ln(t^2+1)}{t^{\frac{4}{3}}},\ e^{\frac{2t}{\sin t}},\ \sqrt{1+t}\,\arctan\Big(\dfrac{1}{|t|}\Big)\Big)$.
Find the limit $\lim_{t\to 0} \vec{r}(t)$.

According to the theory,

$$\lim_{t\to 0}\vec{r}(t) = \left(\lim_{t\to 0}\frac{\ln(t^2+1)}{t^{\frac{4}{3}}},\ \lim_{t\to 0} e^{\frac{2t}{\sin t}},\ \lim_{t\to 0}\sqrt{1+t}\,\arctan\Big(\frac{1}{|t|}\Big)\right).$$

$$\begin{cases} \lim_{t\to 0}\dfrac{\ln(t^2+1)}{t^{\frac{4}{3}}} = \lim_{t\to 0}\dfrac{\frac{2t}{t^2+1}}{\frac{4}{3}t^{\frac{1}{3}}} = \lim_{t\to 0}\dfrac{3}{2}\dfrac{t^{\frac{2}{3}}}{t^2+1} = 0\,; \ {\color{red}(1/2\ \checkmark)} \\ \lim_{t\to 0} e^{\frac{2t}{\sin t}} = e^{\lim_{t\to 0}\frac{2t}{\sin t}} = e^2;\ {\color{red}(1/2\ \checkmark)} \\ \lim_{t\to 0}\sqrt{1+t}\,\arctan\Big(\dfrac{1}{|t|}\Big) = \lim_{t\to 0}\sqrt{1+t}\times\lim_{t\to 0}\arctan\Big(\dfrac{1}{|t|}\Big) = 1\times\dfrac{\pi}{2} = \dfrac{\pi}{2}.\ {\color{red}(1/2\ \checkmark)} \end{cases}$$

Hence, $\lim_{t\to 0}\vec{r}(t) = \left(0,\ e^2,\ \frac{\pi}{2}\right)$. $\color{red}(1/2\ \checkmark)$

[2]

3. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: \left[-\frac{\pi}{2}, \frac{3\pi}{4}\right] \to \mathbb{R}^3$ defined by

$$\vec{r}(t) := \big(2t^2-4,\ e^{-t}\sin(2t),\ e^{-t}\cos(2t)\big), \quad -\frac{\pi}{2} \le t \le \frac{3\pi}{4}.$$

(a) Show that $\mathcal{C}$ is a regular curve.

We want to check on the one hand that the vector function $\vec{r}$ is continuously differentiable, i.e., $\vec{r}$ is differentiable and its derivative $\vec{r}\,'$ is continuous, and on the other hand that $\vec{r}\,'(t) \neq \vec{0}$ for every $t \in \left(-\frac{\pi}{2}, \frac{3\pi}{4}\right)$.
Notice that

$\vec{r}\,'(t) = \big(4t,\ e^{-t}(-\sin(2t)+2\cos(2t)),\ e^{-t}(-\cos(2t)-2\sin(2t))\big)$ $\color{red}(1/2\ \checkmark)$ continuous. $\color{red}(1/2\ \checkmark)$

Now,

$$\vec{r}\,'(t) = (0,0,0) \quad\Rightarrow\quad t = 0.$$

However, $\vec{r}\,'(0) = (0,2,-1) \neq (0,0,0)$. $\color{red}\checkmark$

[2]

(b) Show that $\vec{r}\,''(t) = \big(4,\ e^{-t}(-3\sin(2t)-4\cos(2t)),\ e^{-t}(4\sin(2t)-3\cos(2t))\big)$.

$$\vec{r}\,''(t) = \begin{pmatrix} 4 \\ e^{-t}(\sin(2t)-2\cos(2t)-2\cos(2t)-4\sin(2t)) \\ e^{-t}(\cos(2t)+2\sin(2t)+2\sin(2t)-4\cos(2t)) \end{pmatrix} = \begin{pmatrix} 4 \\ e^{-t}(-3\sin(2t)-4\cos(2t)) \\ e^{-t}(4\sin(2t)-3\cos(2t)) \end{pmatrix} \ {\color{red}\checkmark}$$

[1]

(c) Find the curvature κ of the curve $\mathcal{C}$ at the point $(-4,0,1)$.

We first seek t such that $\vec{r}(t) = (-4,0,1)$. We find for the first componet that $t=0$ and we find indeed that $\vec{r}(0) = (-4,0,1)$. $\color{red}(1/2\ \checkmark)$ The curvature of the curve $\mathcal{C}$ at the point $(-4,0,1)$ is given by the formulas

$$\kappa(0) := \frac{\|\vec{r}\,'(0)\times\vec{r}\,''(0)\|}{\|\vec{r}\,'(0)\|^3} = \frac{\|\vec{T}\,'(0)\|}{\|\vec{r}\,'(0)\|}.\ {\color{red}(1/2\ \checkmark)}$$

It is more convenient to use here the first formula we find $\vec{r}''(0) = (4, -4, -3)$. $(1/2\ \checkmark)$ and

$$\vec{r}'(0) \times \vec{r}''(0) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 2 & -1 \\ 4 & -4 & -3 \end{vmatrix} = (-10\ (1/2\ \checkmark), -4\ (1/2\ \checkmark), -8\ (1/2\ \checkmark)).$$

Thus,

$$\kappa(0) := \frac{\|\vec{r}'(0) \times \vec{r}''(0)\|}{\|\vec{r}'(0)\|^3} = \frac{6\sqrt{5}}{5\sqrt{5}} = \frac{6}{5}\,.\checkmark$$

[4]

(d) Find the unit tangent vector $\vec{T}$ at the point in (c).

Now,

$$\vec{T}(0) = \frac{\vec{r}'(0)}{\|\vec{r}'(0)\|}\ (1/2\ \checkmark) = \frac{1}{\sqrt{5}}(0, 2, -1)\ (1/2\ \checkmark)$$

[1]

(e) Write down the cartesian equation of the normal plane to the curve $\mathcal{C}$ at the point in (c).

The normal plane of the curve $\mathcal{C}$ at the point in (c) has normal vector $\vec{T}\,'(0)$ or $\vec{r}'(0)$ $(1/2\ \checkmark)$ and hence, has cartesian equation

$$\begin{aligned} (x+4, y, z-1)\cdot(0,2,-1) = 0 \quad &\Leftrightarrow \quad 0 \times (x+4) + 2 \times y - 1 \times (z-1) = 0\ (1/2\ \checkmark) \\ &\Leftrightarrow \quad 2y - z = -1\,. \end{aligned}$$

[1]

4. Sketch the domain D of the function f defined by

$$f(x,y) := \sqrt{\ln\Big(\frac{17}{1+x^2+y^2}\Big)} - \ln(y^2 - x - 2)\,.$$

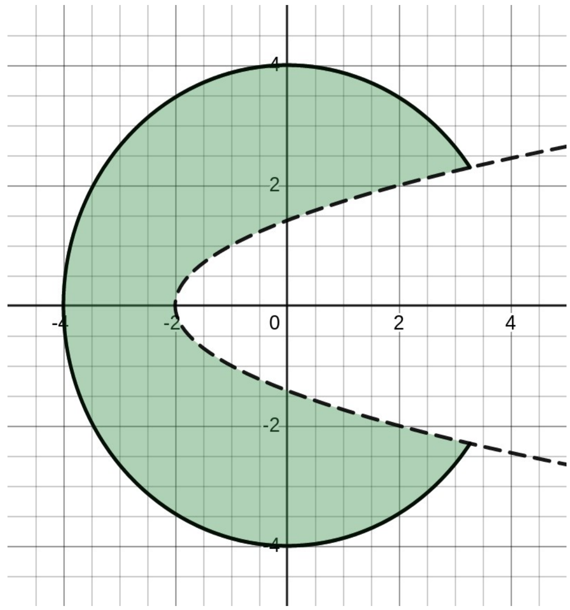
Include all the steps explaining how you got your domain and the sketch!

$f(x,y)$ is defined if and only if

$$\begin{cases} \ln\Big(\dfrac{17}{1+x^2+y^2}\Big) \geq 0,\ (1/2\ \checkmark) \\ y^2 - x - 2 > 0\,.\ (1/2\ \checkmark) \end{cases} \Leftrightarrow \begin{cases} \dfrac{17}{1+x^2+y^2} \geq 1, \\ y^2 - x - 2 > 0\,, \end{cases} \Leftrightarrow \begin{cases} x^2 + y^2 \leq 16, \\ y^2 - x - 2 > 0\,. \end{cases}\ (1/2\ \checkmark)$$

The shaded domain: Correct circle: $(1/2\ \checkmark)$, Correct parabola: $(1/2\ \checkmark)$;
Dotted parabola: $(1/2\ \checkmark)$ Correct shading: $\checkmark$

[4]



5. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) := \begin{cases} \dfrac{xy^2 - x^2}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0)\,. \end{cases}$

(i) Study the continuity of the function f at the point $(0,0)$.

We want to check whether $\lim\limits_{(x,y)\to(0,0)} f(x,y) = f(0,0)$. To this end, we find that

First solution:

$$\begin{cases} \text{Along the } x\text{-axis:} \quad \lim\limits_{x\to 0} f(x,0) = \lim\limits_{x\to 0} \dfrac{-x^2}{|x|} = \lim\limits_{x\to 0} -|x| = 0; \\ \text{Some students would also split } \lim\limits_{x\to 0-} f(x,0) \text{ and } \lim\limits_{x\to 0^+} f(x,0); \\ \text{Along the } y\text{-axis:} \quad \lim\limits_{y\to 0} f(0,y) = \lim\limits_{y\to 0} \dfrac{0}{|y|} = 0; \\ \text{Along the line } x=y\text{:} \quad \lim\limits_{x\to 0} f(x,x) = \lim\limits_{x\to 0} \dfrac{x^3 - x^2}{|x|\sqrt{2}} = \lim\limits_{x\to 0} \dfrac{1}{\sqrt{2}}(x|x| - |x|) = 0\,; \\ \text{Some students would also split } \lim\limits_{x\to 0-} f(x,x) \text{ and } \lim\limits_{x\to 0^+} f(x,x). \end{cases} \checkmark$$

We can consider $L = 0$ as limit candidate. Notice that

$$0 \leq |f(x,y) - 0| \leq \frac{|x|y^2 + x^2}{\sqrt{x^2+y^2}} = \frac{|x|y^2}{\sqrt{x^2+y^2}} + \frac{x^2}{\sqrt{x^2+y^2}} \leq y^2 + |x| \longrightarrow 0\,. \checkmark$$

OR

$$0 \leq |f(x,y) - 0| = \frac{|x||y^2 - |x||}{\sqrt{x^2+y^2}} = \underbrace{\left[\frac{|x|}{\sqrt{x^2+y^2}}\right]}_{\leq 1} |y^2 - |x|| \leq |y^2 - |x|| \longrightarrow 0\,. \checkmark$$

Thus, by the squeeze theorem, $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$ and hence, the functionj f is continuous at $(0,0)$.

Second solution: We first identify the limits along some lines or curves as above to get the tick. ✓ Then we use polar coordinates to show that $\lim_{(x,y)\to(0,0)} f(x,y) = 0$ as follows: $x = r\cos\theta$ and $y = r\sin\theta$ and get

$$\begin{aligned} 0 \le |f(r\cos\theta, r\sin\theta) - 0| &= \frac{|r^3\cos\theta\sin^2\theta - r^2\cos^2\theta|}{r} \\ &\le r^2|\cos\theta|\sin^2\theta + r\cos^2\theta \le r^2 + r \to 0 \text{ as } r \to 0^+ . \checkmark \end{aligned}$$

Hence, $\lim_{r\to 0^+} f(r\cos\theta, r\sin\theta) = 0$ uniformly in θ and therefore, $\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0)$ and hence, the function f is continuous at $(0,0)$.

[2]

(ii) Say whether the partial derivatives $f_x(0,0)$ and $f_y(0,0)$ exist. If they exist, then find them.

For $f_x(0,0$, we find that

$$\frac{f(x,0)-f(0,0)}{x} = \frac{-x^2}{x|x|} \longrightarrow \begin{cases} 1 \;\; (1/2 \checkmark) & \text{as } x \to 0^-, \\ -1 \;\; (1/2 \checkmark) & \text{as } x \to 0^+, \end{cases} \quad \Rightarrow f_x(0,0) \text{ does not exist. } (1/2 \checkmark)$$

For $f_y(0,0$, we find that

$$\frac{f(0,y)-f(0,0)}{y} = \frac{0}{y} \longrightarrow 0 \text{ as } y \to 0 \quad \Rightarrow \quad f_y(0,0) = 0 \,. \; (1/2 \checkmark)$$

[2]

(iii) Explain why the function f is not differentiable at $(0,0)$.

According to the definition of differentiability, if f was differentiable at $(0,0)$, then $f_x(0,0)$ would exist. (1/2 ✓) Therefore, f is not differentiable at $(0,0)$. (1/2 ✓)

[1]

Section B **Full Mark: 10**

1. The curve $\mathcal{C}$ described by the vector function $\vec{r}(t) := \left(\frac{1-(\ln t)^2}{1+(\ln t)^2}, e^{2t}, \frac{2(\ln t)}{1+(\ln t)^2}\right)$ with $t > 1$, lies

(a) on the surface $\ln(\ln y) - xz = 0$; (b) on the cylinder $x^2 + z^2 = 1$; (c) in the set $\{x > 1\}$;

(d) on the sphere $x^2 + y^2 + z^2 = 1$; (e) on the cone $y^2 = x^2 + z^2$.

Solution: (b)

[2]

2. The curve $\mathcal{C}$ described by the vector function $\vec{r}(t) := \left(2\sin^2(t), t, \frac{3}{2}\sin(2t)\right)$ with $-\pi \le t \le \frac{\pi}{4}$ has length $L(\mathcal{C})$ equal to

(a) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{4 + t^2 + 5\cos^2(2t)}\, dt$; (b) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{5\cos^2(2t)}\, dt$; (c) $\frac{5\pi\sqrt{5}}{4}$;

(d) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{4\sin^2(2t) + 1 + 9\cos^2(2t))}\, dt$; (e) $\int_{-\pi}^{\frac{\pi}{4}} \sqrt{4\sin^4(t) + t^2 + \frac{9}{4}\sin^2(2t)}\, dt$.

Solution: (d)

[2]

3. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) := \displaystyle\int_x^y \sin(t^2)dt$. Then its gradient i.e., the vector having components its partial derivatives, at the point $\left(\frac{\sqrt{\pi}}{2}, \sqrt{\frac{3\pi}{2}}\right)$ is equal to:

(a) $\left(\frac{\sqrt{2}}{2}, -1\right)$; (b) $\left(\frac{\sqrt{2}}{2}, 0\right)$; (c) $\left(-\frac{\sqrt{2}}{2}, 1\right)$; (d) $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$; (e) $\left(-\frac{\sqrt{2}}{2}, -1\right)$.

Solution: (e)

[2]

4. The vector function $\vec{r}$ with expression $\vec{r}(t) := \left(\ln(4-t^2), \frac{1}{\sqrt{2-e^t}}, \frac{3t}{2t^2-4t+3}\right)$ has domain $\mathrm{Dom}(\vec{r})$ equal to the interval:

(a) $(-2, \ln 2)$; (b) $(\ln 2,\ 2)$; (c) $(\ln 2,\ e^4)$; (d) $(-4, \ln 2)$; (e) $[-2, \ln 2]$.

Solution: (a)

[2]

5. The level curve of the function $f(x,y) = \sin(xy) - \cos(xy)$ passing through the point $(-1, \pi)$

(a) does intersect the y-axis;
(b) also passes through the point $\left(-\sqrt{\pi}, \frac{\sqrt{\pi}}{2}\right)$;
(c) has cartesian equation $\cos(xy) - \sin(xy) = -1$;
(d) also passes through the point $\left(\frac{\sqrt{\pi}}{2}, \frac{\sqrt{\pi}}{2}\right)$;
(e) intersects the x-axis at the point $\left(-\frac{\pi}{2}, 1\right)$.

Solution: (c)

[2]